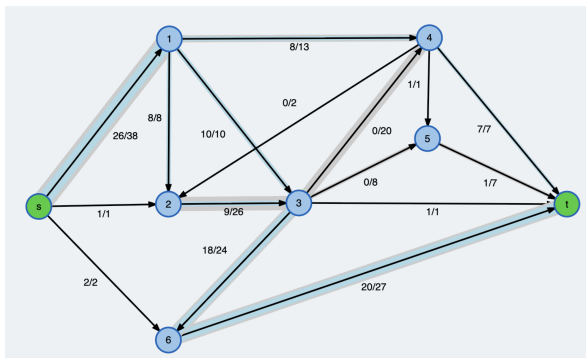
CS 344: Design and Analysis of Computer Algorithms **Rutgers: Spring 2026**

Network Flow

1. Find the residual graph of the following flow network. Run the Ford-Fulkerson algorithm, find the maximum flow and a minimum cut of the graph.



2. Suppose you are given a directed acyclic graph G with a single source vertex s. Describe an algorithm to determine whether G contains a spanning binary tree. Your algorithm is looking for a spanning tree T of G, such that every vertex in G has at most two outgoing edges in T and every vertex of G except s has exactly one incoming edge in T.

3. You are coordinating the grading operation for a massive end-of-semester final exam. You have a team of N Teaching Assistants (graders) and the exam consists of M different questions.You must assign graders to questions subject to the following constraints: Each grader i has specialized knowledge and is only qualified to grade a specific subset of the M questions.To get the grading done by the deadline, each exam question j must have exactly C_j graders assigned to it. No single grader can be assigned to more than W questions in total. The M questions are grouped into K overarching algorithmic topics. To prevent graders from burning out on one specific type of rigorous proof, a grader can be assigned to a maximum of L questions within any single topic category k. Formulate this as a network flow problem to determine if a valid grading assignment exists that satisfies all requirements, and if so, how to find it.

4. A university department needs to form thesis defense committees for M graduating PhD students. There are N faculty members available to serve.You must assign faculty to students subject to the following rules. Faculty member i can only serve on student j's committee if their research areas overlap (you are given this list of valid pairs). Every student j must have exactly 3 faculty members on their committee. Faculty member i has limited bandwidth and can serve on at most U_i committees.To prevent junior faculty from taking on all the administrative burden, departmental policy dictates that every faculty member i must serve on at least L_i committees. Formulate this as a network flow problem to determine if a valid committee assignment exists that satisfies all rules simultaneously.

5. Let $G = (U, V, E)$ be a bipartite graph where $U = \{s_1, s_2, s_3, s_4, s_5\}$ represents a set of students and $V = \{p_1, p_2, p_3, p_4, p_5\}$ represents a set of projects.The Preference Table is the following :$s_1 \to \{p_2, p_5\}$, $s_2 \to \{p_1, p_3\}$, $s_3 \to \{p_5\}$, $s_4 \to \{p_2, p_5\}$, $s_5 \to \{p_3, p_4\}$ Run the augmenting path flow algorithm to find a maximum matching and find a set of students such that $|N(S)| < |S|$ if perfect matching doesn't exist. (Hall's Marriage theorem guarantees such a set exists if there is no perfect matching)